SUPERSEDING:
See Section 6

# MILITARY SPECIFICATION

## SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, GERMANIUM
## TYPES 2N416, 2N417

## 1. SCOPE

1.1 Scope.- This specification covers the detail requirements for germanium, PNP, transistors for use as low-power devices in compatible electronic-equipment circuits. (See 6.2 herein.)

1.2 Outline and dimensions.- See Figure 1 herein. (TO-5)

1.3 Maximum ratings.- (At $T_A$=+25°C, unless otherwise specified):

| | $P_T$ 1/ | $I_{CBO}$ | $I_{EBO}$ | $V_{CBO}$ | $V_{EBO}$ | $T_{stg}$ | $T_{(oper)}$ |
|---|---|---|---|---|---|---|---|
| | mW | uAdc | uAdc | Vdc | Vdc | °C | °C |
| 2N416 | 150 | -25 | -5 | -30 | -12 | -65 to +100 | -65 to +100 |
| 2N417 | 150 | -25 | -5 | -30 | -12 | -65 to +100 | -65 to +100 |

1/ For $T_A$>+25°C, derate at 2.0 mW/°C.

1.4 Particular electrical characteristics.- (At $T_A$=+25°C, unless otherwise specified.):

| | $h_{fe}$ at:f= 1 KHz $V_{CB}$=-6 Vdc $I_E$=-1 mAdc | | $V_{CE}$ at: $V_{BE}$= 1 Vdc $I_C$=-25 uAdc | | $r_b$ at:f=1 KHz $V_{CB}$=-6 Vdc $I_E$=-1 mAdc | | $f_{hfb}$ at: $V_{CB}$=-6 Vdc $I_E$=-1 mAdc | | $C_{obo}$ at: 1/ $V_{CB}$=-6 Vdc $I_E$= 0 | |
|---|---|---|---|---|---|---|---|---|---|---|
| | --- | | Vdc | | ohms | | MHz | | pf | |
| | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max |
| 2N416 | 45 | 220 | -12 | --- | --- | 150 | 5 | --- | --- | 20 |
| 2N417 | 60 | 260 | -12 | --- | --- | 170 | 15 | --- | --- | 20 |

1/ f ≥ 0.1≤ 1.0 MHz.

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS
MILITARY
MIL-S-19500 Semiconductor Devices, General Specification For

STANDARDS
MILITARY

MIL-STD-750 Test Methods for Semiconductor Devices

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Requirements.- Requirements for the transistors shall be in accordance with Specification MIL-S-19500 and as otherwise specified herein.

3.2 Abbreviations and symbols.- The abbreviations and symbols used herein are defined in Specification MIL-S-19500.

3.3 Design and construction.- The transistor shall be of the design, construction, and physical dimensions specified on Figure 1. (See 3.3.2 herein.)

3.3.1 Terminal arrangement.- The terminal arrangement on the transistor shall be as indicated in Figure 1.

3.3.2 Terminal-lead length.- Terminal-lead length(s) other than that specified in Figure 1 may be furnished under contract or order (see 6.3 herein) where the devices covered herein are required directly for particular equipment-circuit installation. Where such other lead lengths are required and provided, it shall not be construed as affecting adversely the Qualified-product status of the device, or applicable JAN marking.

3.3.3 Operating position.- The transistor shall be capable of proper operation in any position.



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3.4 Performance characteristics.- The transistor performance characteristics shall be as specified in Tables I, II, and III herein. Except where specifically differentiated for respective transistor types (see 1.3, 1.4, and Tables I, II, and III herein), the performance requirements, including characteristics, ratings, and test conditions, apply equally to both transistor types covered herein.

3.5 Marking.- Except as otherwise specified herein, marking shall be in accordance with Specification MIL-S-19500. The "manufacturer's identification" may, at option of the manufacturer, be omitted from being marked on the body of the semiconductor device covered herein.

4. QUALITY ASSURANCE PROVISIONS.

4.1 General.- Except as otherwise specified herein, the responsibility for inspection, general procedures for acceptance, classification of inspection, and inspection conditions and methods of test shall be in accordance with Specification MIL-S-19500, Quality Assurance Provisions.

4.2 Qualification and acceptance inspection.- Qualification and Quality Conformance inspection shall be in accordance with Specification MIL-S-19500, Quality Assurance Provisions, and as otherwise specified herein (see 4.2.2 herein). Groups A, B, and C inspection shall consist of the examinations and tests specified in Tables I, II, and III, respectively, herein. Quality Conformance inspection shall include inspection of Preparation for Delivery (see 5.1 herein).

4.2.1 Special Group A criteria for Qualification inspection.- For qualification inspection, only 10 failures will be permitted for all Group A tests combined. Hereto, where the manufacturer deems that the data in paragraph 4.4.2 of Specification MIL-S-19500 are invokable, notification shall be made to the Chief, Components and Materials Standardization Division, Procurement and Production Directorate, Fort Monmouth, New Jersey 07703, Attention: AMSEL-PP-EM-2.

4.2.2 Group B-Group C life test samples.- Samples that have been subjected to Group B, 340-hour life test may be continued on test for 1000 hours in order to satisfy Group C life test requirements. These samples shall be predesignated, and shall remain subject to the Group C 1000-hour acceptance evaluation after they have passed the Group B, 340-hour acceptance criteria. The cumulative total of failures found during 340-hour test and during the subsequent interval up to 1000 hours on these samples shall be computed for 1000-hour acceptance criteria.

4.2.3 Group C testing.- Unless otherwise specified, Group C tests shall be performed on the initial lot and thereafter on a lot every 6 months. (See Table III herein.) The contractor shall, throughout the course of a contract or order, permit the Government representative to scrutinize all test data and findings covering manufacturer's test program on Group C characteristics and parameters for the product concerned. Upon determination by the Government inspector (in advance of Group C, 6-month, test results) that Group C parameters are not being adequately met, the Government inspector may require lot-by-lot inspection, normally for a minimum of 3 consecutive lots, to be performed for required Group C tests.



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4.2.4 Disposition of sample units.- Sample units that have been subjected to Group B, Subgroups 2, 4, and 5 tests shall not be delivered on the contract or order. Sample units that have been subjected to and have passed Group B, Subgroups 1, 3, 6, and 7 tests, and Group C, Subgroup 1 and 2 tests (these tests to be considered non-destructive), may be delivered on the contract or order provided that, after Group B inspection is terminated, those sample units are subjected to and pass Group A inspection. Defective units from any sample group that may have passed group inspection shall not be delivered on the contract or order until the defect(s) has been remedied to the satisfaction of the Government.

4.3 Particular examination and test requirements.-

4.3.1 Mechanical damage resulting from tests.- Except for intentionally deforming, mutilating, or dismembering mechanical-stress tests to which samples are subjected, there shall be no evidence of mechanical damage fo any sample unit as a result of any of the Groups A, B, or C tests.

4.3.2 Seal (leak-rate) test.- The transistors shall be subjected to hermetic-seal test(s) in accordance with requirements of Method 1071 in Standard MIL-STD-750 except that the following test conditions therein shall apply hereto:

a. Fine-leak test: per Test Condition G or H. Where Condition H is employed, a maximum leakage rate of $1 \times 10^{-7}$ atm/cc/sec shall be effective for acceptance.

b. Gross-leak test: per Test Condition A, C, D; or F1



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Table I. Group A inspection.

| Test Method per MIL-STD-750 | Examination or test 1/ | Conditions | LTPD | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|
| | Subgroup 1 | | 10 | | | | |
| 2071 | Visual and mechanical examination | --- | | --- | --- | --- | --- |
| | Subgroup 2 | | 7 | | | | |
| 3036 | Collector-base cutoff current | Bias Cond. D $V_{CB}$=-12 Vdc | | $I_{CBO}$ | --- | -5 | uAdc |
| 3036 | Collector-base cutoff current | Bias Cond. D $V_{CB}$=-30 Vdc | | $I_{CBO}$ | --- | -25 | uAdc |
| 3061 | Emitter-base cutoff current | Bias Cond. D $V_{EB}$=-12 Vdc | | $I_{EBO}$ | --- | -5 | uAdc |
| 3030 | Collector-emitter voltage | $V_{BE}$= 1 Vdc $I_C$=-25 uAdc | | $V_{CE}$ | -12 | --- | Vdc |
| | Subgroup 3 | | 7 | | | | |
| 3206 | Small-signal short-circuit forward-current transfer ratio: | $V_{CB}$=-6 Vdc $I_E$= 1 mAdc f= 1 KHz | | | | | |
| | 2N416 | | | $h_{fe}$ | 45 | 220 | --- |
| | 2N417 | | | $h_{fe}$ | 60 | 260 | --- |
| 3301 | Small-signal short-circuit forward-current transfer ratio cutoff frequency: | $V_{CB}$=-6 Vdc $I_E$= 1 mAdc | | | | | |
| | 2N416 | | | $f_{hfb}$ | 5 | --- | MHz |
| | 2N417 | | | $f_{hfb}$ | 15 | --- | MHz |



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Table I. Group A inspection.-(Cont'd).

| Test Method per MIL-STD-750 | Examination or test 1/ | Conditions | LTPD | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|
| | Subgroup 3-(Cont'd) | | | | | | |
| 3236 | Output capacitance (open circuit) | $V_{CB}$=-6 Vdc<br>$I_E$= 0<br>f≥0.1≤1.0 MHz | | $C_{obo}$ | --- | 20 | pf |
| 3266 | Base spreading resistance: | $V_{CB}$=-6 Vdc<br>$I_E$= 1 mAdc | | | | | |
| | 2N416 | | | $r_b$ | --- | 150 | ohms |
| | 2N417 | | | $r_b$ | --- | 170 | ohms |

1/ See 3.4 herein.



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Table II. Group B inspection.

| Test Method per MIL-STD-750 | Examination or test 1/ | Conditions | LTPD | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|
| | Subgroup 1. | | 20 | | | | |
| 2066 | Physical dimensions | --- | | --- | --- | --- | --- |
| | Subgroup 2 | | 15 | | | | |
| 2026 | Solderability | --- | | --- | --- | --- | --- |
| 1051 | Temperature cycling | Test Cond. B except $T_{(high)}$= +100°C | | --- | --- | --- | --- |
| 1056 | Thermal shock (glass strain) | Test Cond. A | | --- | --- | --- | --- |
| 2/ | Seal (leak rate) | --- | | --- | --- | --- | --- |
| 1021 | Moisture resistance | --- | | --- | --- | --- | --- |
| | End-point tests: | | | | | | |
| 3036 | Collector-to-base cutoff current | Bias Cond. D $V_{CB}$=-30 Vdc | | $I_{CBO}$ | --- | 25 | uAdc |
| 3206 | Small-signal short-circuit forward-current transfer ratio: | $V_{CB}$=-6 Vdc $I_E$= 1 mAdc $f$= 1 KHz | | | | | |
| | 2N416 | | | $h_{fe}$ | 45 | 220 | --- |
| | 2N417 | | | $h_{fe}$ | 60 | 260 | --- |



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Table II. Group B inspection-(Cont'd).

| Test Method per MIL-STD-750 | Examination or test 1/ | Conditions | LTPD | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|
| | Subgroup 3 | | 15 | | | | |
| 2016 | Shock | Non-operating 1500 G 5 blows of 0.5 msec ea. in orientations X1, Y1, Y2, Z1 (total=20 blows) | | --- | --- | --- | --- |
| 2056 | Vibration, variable frequency | --- | | --- | --- | --- | --- |
| 2006 | Constant acceleration | 20,000 G Orientations X1, Y1, Y2, Z1 | | --- | --- | --- | --- |
| | End-point tests: Same as for Subgroup 2 above | | | | | | |
| | Subgroup 4 | | 20 | | | | |
| 1041 | Salt atmosphere (corrosion) | 3/ | | --- | --- | --- | --- |
| | Subgroup 5 | | 20 | | | | |
| 2036 | Terminal strength: Lead fatigue | 3/ Test Cond. E | | --- | --- | --- | --- |



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Table II. Group B inspection-(Cont'd).

| Test Method per MIL-STD-750 | Examination or test 1/ | Conditions | LTPD | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|
| | Subgroup 6 | | 7 | | | | |
| 1031 | High-temperature life (non-operating) | $T_{stg}$=+100°C t=340 hrs;4/ | | --- | --- | --- | --- |
| | End-point tests: | | | | | | |
| 3036 | Collector-base cut-off current | Bias Cond. D $V_{CB}$=-30 Vdc | | $I_{CBO}$ | --- | 40 | uAdc |
| 3206 | Small-signal short-circuit forward-current transfer ratio: | $V_{CB}$=-6 Vdc $I_E$=-1 mAdc f= 1 KHz | | | | | |
| | 2N416 | | | $h_{fe}$ | 35 | 250 | --- |
| | 2N417 | | | $h_{fe}$ | 50 | 300 | --- |
| | Subgroup 7 | | 7 | | | | |
| 1026 | Steady-state operation life | $T_A$=+25°C $V_{CB}$=-30 Vdc $P_T$=150 mW t=340hrs;4/ | | --- | --- | --- | --- |
| | End-point tests: Same as for Subgroup 6 above | | | | | | |

1/ See 3.4 herein.

2/ See 4.3.2 herein.

3/ Electrical rejects from the same lot under evaluation may be used for this test.

4/ See 4.2.2.herein.



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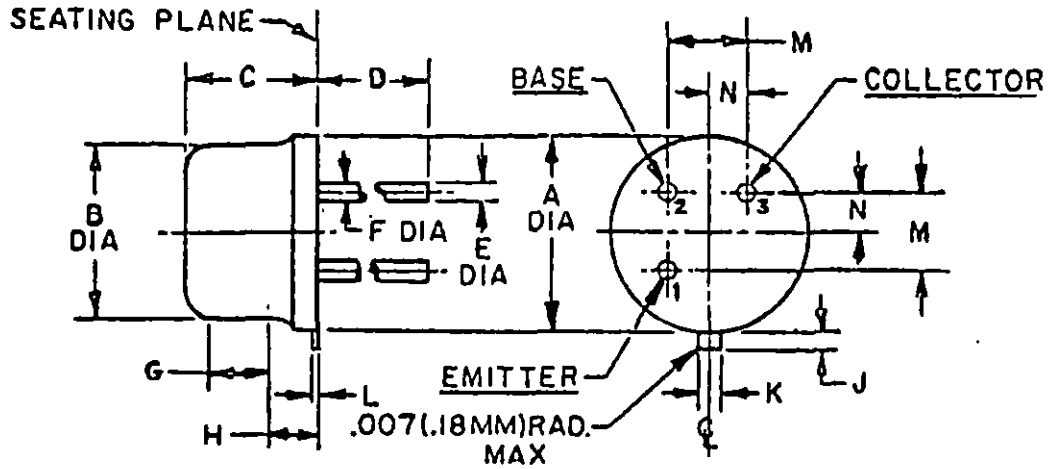
Table III. Group C inspection.[1/]

| Test Method per MIL-STD-750 | Examination or test [2/] | Conditions | LTPD | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|
| | Subgroup 1 | | $\lambda$=7 | | | | |
| 1031 | High temperature life (non-operating) | $T_{stg}$=+100°C<br>t=1000 hrs;[3/] | | --- | --- | --- | --- |
| | End-point tests: | | | | | | |
| 3036 | Collector-base cut-off current | Bias Cond. D<br>$V_{CB}$=-30 Vdc | | $I_{CBO}$ | --- | 40 | uAdc |
| 3206 | Small-signal short-circuit forward-current transfer ratio: | $V_{CB}$=-6 Vdc<br>$I_E$=-1 mAdc<br>f= 1 KHz | | | | | |
| | 2N416 | | | $h_{fe}$ | 35 | 250 | --- |
| | 2N417 | | | $h_{fe}$ | 50 | 300 | --- |
| | Subgroup 2 | | $\lambda$=7 | | | | |
| 1026 | Steady-state operation life | $T_A$=+25°C<br>$V_{CB}$=-30 Vdc<br>$P_T$=150 mW<br>t=1000 hrs;[3/] | | | | | |
| | End-point tests: Same as for Subgroup 1 above | | | | | | |

[1/] See 4.2.3 herein. [2/] See 3.4 herein. [3/] See 4.2.2 herein.



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| LTR | DIMENSIONS | | | | NOTES |
|---|---|---|---|---|---|
| | INCHES | | MILLIMETERS | | |
| | MIN | MAX | MIN | MAX | |
| A | .335 | .370 | 8.51 | 9.40 | |
| B | .305 | .335 | 7.75 | 8.51 | |
| C | .240 | .260 | 6.10 | 6.60 | |
| D | 1.500 | 1.750 | 38.10 | 44.45 | 7,9 |
| E | .016 | .021 | .41 | .53 | 2,9 |
| F | .016 | .019 | .41 | .48 | 3,9 |
| G | .100 | --- | 2.54 | --- | 4 |
| H | --- | --- | --- | --- | 5 |
| J | .029 | .045 | .74 | 1.14 | 8 |
| K | .028 | .034 | .71 | .86 | |
| L | .009 | .125 | .23 | 3.18 | |
| M | .1414 Nom | | 3.59 Nom | | 6 |
| N | .0707 Nom | | 1.80 Nom | | 6 |

1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch =25.4mm.
2. Measured in the zone beyond .250 (6.35mm) from the seating plane.
3. Measured in the zone .050 (1.27mm) and .250(6.35mm) from the seating plane.
4. Variations on Dim.B in this zone shall not exceed .010 (.25mm).
5. Outline in this zone is not controlled.
6. When measured in a gaging plane .054 +.001 (1.37mm +.03mm) below the seating plane of the transistor, max dia leads shall be within .007 (.18mm) of their true location relative to a maximum width tab. Smaller dia leads shall fall within the outline of the max dia lead tolerance.
7. No leads internally connected to case.
8. Measured from the maximum diameter of the actual device.
9. All 3 leads.

Figure 1. Outline and dimensions.



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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery.- Preparation for delivery, and the inspection of preparation for delivery, shall be in accordance with Specification MIL-S-19500.

6. NOTES

6.1 Notes.- The notes included in Specification MIL-S-19500, with the following additions or exceptions, are applicable to this document.

6.2 Application guidance and supersession data.-

a. The transistors conforming to requirements of this document issue are recommended as ready replacements (having superior-controlled characteristics) for the transistors covered by the previous issue of, respectively, this document and the additional document listed below:

MIL-T-19500/56A(Sig C), 3 February 1959--Transistor, Type 2N416
MIL-T-19500/57A(Sig C), 3 February 1959--Transistor, Type 2N417

b. To insure proper equipment-circuit application, particular attention should be given to the differential cutoff-frequency and gain characteristics and ratings pertinent to the individual transistor types covered herein.

6.3 Ordering data.-

a. Terminal-lead length: See 3.3.2 herein.

6.4 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List (QPL)-19500, whether or not such products have actually been so listed by that date. Information pertaining to qualification of products covered by this specification should be requested from the Commanding General, U. S. Army Electronics Command, Fort Monmouth, New Jersey 07703, Attention: AMSEL-PP-EM-2.

6.5 Revision (document) changes.- Revision-letter symbols are not used in this document revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army-EL

Preparing activity:
Army-EL

Project No. 5961-A216



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions - Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| | |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION (Mark one) |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER (Specify): _____ |
| b. ADDRESS (Street, City, State, ZIP Code) | |
| | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER (Last, First, MI) – Optional | b. WORK TELEPHONE NUMBER (Include Area Code) – Optional |
|---|---|
| c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional | 8. DATE OF SUBMISSION (YYMMDD) |
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